

Problem 3.13

Suppose $a(t) = 1.12^t - 0.05\sqrt{t}$.

- (a) How much interest will be earned during the 5th year on an initial investment of \$12?
- (b) What is the effective annual interest rate during the 5th year?

Solution

Let $A(0) = 12$. Since

$$a(t) = \frac{A(t)}{A(0)},$$

we have

$$A(t) = 12 a(t).$$

First compute

$$a(5) = 1.12^5 - 0.05\sqrt{5} \approx 1.6505.$$

Thus,

$$A(5) = 12(1.6505) = 19.806.$$

Now compute

$$a(4) = 1.12^4 - 0.05\sqrt{4} = 1.12^4 - 0.1 \approx 1.4735.$$

Hence,

$$A(4) = 12(1.4735) = 17.6822.$$

- (a) The interest earned during the 5th year is

$$I_5 = A(5) - A(4).$$

Therefore,

$$I_5 = 19.806 - 17.6822 = 2.1238.$$

- (b) The effective annual interest rate during the 5th year is

$$i_5 = \frac{A(5) - A(4)}{A(4)}.$$

So,

$$i_5 = \frac{19.806 - 17.6822}{17.6822} \approx 0.1201.$$

Therefore,

$$i_5 \approx 12.01\%.$$